Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** DRBENCHMARK* | ***Context:** EU Pharmaceutical Regulatory NLP — French*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology shows fundamental misalignment with the deployment, which the elicitation marked HIGH priority. Two critical violations are present. (1) STS scoring uses general semantic proximity on a 0–5 scale [Q41, Q54], whereas the deployment requires regulatory-equivalence scoring sensitive to dose-threshold and population-qualifier micro-differences (elicitation A3). The dataset analysis empirically demonstrates the calibration is incompatible: a 2× quantitative difference scores 4.0, a 14× difference scores 3.75, and a 60× PSA difference scores 3.0 (CLISTER-D12, CLISTER-D14, CLISTER-D13); legally significant additions of specific opioid names or 'sous stricte surveillance médicale' qualifiers are not penalized in DEFT2020 (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20). (2) NER label categories are calibrated to clinical/UMLS taxonomies that conflate distinct regulatory entity types: QUAERO and Mantra-GSC's CHEM tag merges INN, brand name, and excipient (QUAERO-D10, QUAERO-D11, MANTRAGSC-D5, MANTRAGSC-D14); ATC codes and EMA administrative identifiers receive O tag (QUAERO-D12); contraindication qualifiers are tagged by entity type rather than as regulatory contraindication structure (QUAERO-D16, MANTRAGSC-D4). Web search confirms no French NLP benchmark with INN/ATC/excipient/contraindication-specific schemas exists [WEB-11, WEB-12]. The leading model also does not excel at MLC or STS [Q82] — the dimensions on which the deployment most depends. This is a structural validity violation.

ID	Evidence
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)
Q82	"Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS." (p.8)
MANTRAGSC-D4	Vous ne devez pas prendre Pramipexole Teva si vous allaitez. — Contraindication in patient leaflet register; condition tagged DISO not as contraindication type
MANTRAGSC-D5	1 ml de solution contient 40 microgrammes de travoprost et 5 mg de timolol (sous forme de maléate de timolol) — INN and salt excipient form both tagged CHEM; no INN/excipient distinction
QUAERO-D10	Contient également: mannitol, hydroxyde de sodium. — Excipients tagged identically to INNs (CHEM), preventing entity sub-class distinction
QUAERO-D11	Refludan 50 mg en poudre pour solution injectable… le mannitol (E 421) et l'hydroxyde de sodium pour l'ajustement du pH. — Excipient same tag as INN, conflating distinct regulatory entity types
CLISTER-D12	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — 2× quantitative difference scored 4.0 (very similar)
QUAERO-D12	EMEA/H/C/122 — EMA product number tagged O (no entity), missing key regulatory identifier class
CLISTER-D13	Le taux de PSA était de 218 ng/ml (normale ≤ 4ng/ml)." / "Le dosage de PSA était de 13327 ng/ml (normal : < 4 ng/ml). — 60× magnitude difference scored 3.0
CLISTER-D14	Le taux de PSA était de 4,49 ng/ml." / "Le taux de PSA total était à 66,68 ng/ml. — 14× numeric discrepancy scored near-similar
MANTRAGSC-D14	Renagel 800 mg sevelamer — Brand name and INN both CHEM; no proprietary/INN label distinction critical for regulatory workflows

DEFT2020-D16	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked
QUAERO-D16	Par conséquent, Refludan ne doit pas être administré à la femme enceinte ou qui allaite. — Contraindication clause with pregnancy qualifier — UMLS tags entity type but not regulatory contraindication structure
DEFT2020-D17	Chez les patients insuffisamment équilibrés par glimepiride arrow à la dose maximale, un traitement par l'insuline peut être associé si nécessaire. — "sous stricte surveillance médicale" absent in source; critical qualifier not flagged in score
DEFT2020-D20	L'absorption de boissons alcoolisées est fortement déconseillée pendant le traitement (potentialisation réciproque). — Mechanism qualifier "(potentialisation réciproque)" dropped in cible; not flagged as regulatory deviation by score
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-12	Bayer et al. 2021 — used pharmacovigilance evaluators for ADE extraction but not regulatory equivalence scoring; closest comparator confirms gap (https://link.springer.com/article/10.1007/s40264-020-00996-3)

Strengths

- Pharmaceutical entity types are partially captured: PxCorpus has 38 NER classes including DRUG, DOSE, MODE, FREQUENCY, DURATION [Q44, Q108] (PXCORPUS-D1, PXCORPUS-D2)
- DEFT2021 NER includes SUBSTANCE, DOSAGE, MODE, DURATION as deployment-relevant entity classes (DEFT2021-D3, DEFT2021-D4)
- Replace intent class in PxCorpus directly captures dosage correction acts analogous to compliance flagging (PXCORPUS-D6, PXCORPUS-D7)
- STS exists as an output type with continuous 0–5 scoring and Spearman/EDRM evaluation [Q54], even if calibration mismatches the deployment

ID	Evidence
Q44	“PxCorpus (Kocabiyikoglu et al., 2022) is a spoken language understanding dataset in the domain of medical drug prescription transcripts. It includes 4 hours (1,981 recordings) of transcribed and annotated dialogues focused on drug prescriptions. The recordings were manually transcribed and semantically annotated. The first task involves classifying the textual utterances into one of the 4 intent classes (prescribe, replace, negate, none). The second task is a NER task where each word in a sequence is classified into one of 38 classes, such as drug, dose, or mode (more detail in Appendix B.9).” (p.4)
Q54	“For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020).” (p.6)
Q108	“Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE” (p.14)
PXCORPUS-D1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Full prescription structure with drug name, dose, form, frequency, indication, duration
PXCORPUS-D2	ténofovir 245 milligrammes en comprimés à prendre après les repas 1 comprimé le soir pendant 2 semaines puis stop — INN drug name with dosage, route instruction, and duration
DEFT2021-D3	rifampicine – isoniazide – pyrazinamide pendant 6 mois — Drug names tagged as SUBSTANCE with duration — partial INN coverage
DEFT2021-D4	Hydrocortisone 300 mg IV... Hydroxyzine 25 mg par voie orale toutes les 6 heures — Drug + dose + route + frequency tagged; matches deployment posology targets
PXCORPUS-D6	attention il s'agit de 20 milligrammes et pas 10 milligrammes — Explicit dosage correction; directly relevant to dose-discrepancy detection
PXCORPUS-D7	remplacer 1 comprimé tous les jours par 1 comprimé en cas d'anxiété — Frequency/indication substitution captured by replace intent class

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance

OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories reviewed: NER schemas span UMLS Semantic Groups (10–11 classes in QUAERO/Mantra-GSC) [Q35, Q103, Q106], clinical entity layers (CLINENTITY/EVENT/ACTOR/etc. in E3C) [Q104], 38-class pharmaceutical schema (PxCorpus) [Q108], 13-type fine-grained NER (DEFT2021) [Q28], and classification schemas (MeSH-C 23 axes [Q27], 12 specialties [Q105], 22 ICD-10 chapters [Q45]). STS uses 0–5 continuous scale [Q41].

OO-2: Missing categories specific to the regional regulatory context: (a) INN as distinct from brand name and excipient — absent across all NER schemas (QUAERO-D7, QUAERO-D10, MANTRAGSC-D5); (b) ATC code as entity class — absent (QUAERO-D12), and confirmed by [WEB-11] as field-level gap; (c) excipient name as distinct regulatory entity — absent (QUAERO-D11, MANTRAGSC-D8); (d) EMA-templated posology structure — absent at template-fidelity level; (e) contraindication qualifier as regulatory construct — absent (QUAERO-D16, MANTRAGSC-D4); (f) regulatory-equivalence STS scale — absent; CLISTER calibration is incompatible (CLISTER-D12, CLISTER-D13, CLISTER-D14).

OO-3: Categories encoding non-regional or non-deployment values: ICD-10 chapter classification (DiaMED) reflects clinical disease taxonomy not regulatory document structure; MeSH-C axes (DEFT2021) reflect biomedical literature indexing not regulatory data attributes; UMLS Semantic Groups [Q35] are research-oriented and conflate regulatory-distinct chemicals into single CHEM class.

OO-4: Stakeholder-driven taxonomy redesign is required: regulatory affairs specialists would need to define a regulatory entity schema (INN/ATC/excipient/posology/contraindication-qualifier) and a regulatory-equivalence STS rubric, neither of which currently exists in DrBenchmark or, per web search, in any French NLP benchmark [WEB-11, WEB-12].

OO-5: Taxonomy issues harming structural and content validity: STS scoring rule fundamentally mismatches the deployment's regulatory-equivalence decision rule (CLISTER-D12, CLISTER-D13, CLISTER-D14, DEFT2020-D16, DEFT2020-D17, DEFT2020-D20); NER label space conflates regulatorily distinct entity types (QUAERO-D10, QUAERO-D11, MANTRAGSC-D14); regulatory-specific entity types are absent from all schemas (QUAERO-D12).

ID	Evidence
Q27	“The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH).” (p.3)
Q28	“The second task involves fine-grained information extraction for 13 types of entities (more detail in Appendix B.7).” (p.3)
Q35	“10 entity categories corresponding to the UMLS Semantic Groups (Lindberg et al., 1993) were annotated (more detail in Appendix B.3).” (p.3)
Q41	“CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided.” (p.4)
Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)
Q54	“For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020).” (p.6)
Q82	“Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS.” (p.8)
Q103	“O, GEOG, PHEN, DISO, ANAT, OBJC, PHYS, PROC, DEVI, CHEM and LIVB” (p.14)
Q104	“Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML” (p.14)

Q105	“microbiology, etiology, virology, physiology, immunology, parasitology, genetics, chemistry, veterinary, surgery, pharmacology and psychology” (p.14)
Q106	“Medline: ANAT, PROC, CHEM, PHYS, GEOG, DEVI, LIVB, OBJC, DISO, PHEN and O. EMEA and Patents: ANAT, PROC, CHEM, PHYS, DEVI, LIVB, OBJC, DISO, PHEN and O.” (p.14)
Q108	“Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE” (p.14)
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QUAERO-D7	EMEA/H/C/122 REFLUDAN… Son principe actif est la lépirudine. — Brand name and INN both tagged CHEM, demonstrates INN extraction capability
MANTRAGSC-D8	L' utilisation selon la revendication 3, où le laxatif osmotique est du glycol de polyéthylène 3350. — Excipient/drug substance in patent claim; CHEM tag does not distinguish excipient from INN
QUAERO-D10	Contient également: mannitol, hydroxyde de sodium. — Excipients tagged identically to INNs (CHEM), preventing entity sub-class distinction
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Information Gaps

- No quantification of how often clinical and regulatory NER label decisions would diverge on the same span (e.g., would CHEM-tagged 'natalizumab' match an INN-class regulatory annotation in a controlled comparison).

Requires Expert Verification

- Whether DEFT2021's SUBSTANCE/DOSAGE/MODE/DURATION schema, when applied to actual SmPC text, would yield acceptable regulatory-grade extraction or would require relabeling.
- Whether the EDRM metric [Q54] could be re-purposed for regulatory-equivalence scoring with a new annotation rubric.

Remediation

Gap 1: STS scoring is calibrated for general semantic proximity and demonstrably tolerates regulatorily significant differences (CLISTER 60× quantitative divergence scores 3.0; DEFT2020 specific drug-name additions score 4.0).

Recommendation: Develop a complementary regulatory-equivalence STS rubric with explicit penalties for dose-threshold changes, population-qualifier changes, and INN substitutions. Annotate a held-out evaluation set of EMA SmPC safety-warning sentence pairs against this rubric and report Spearman correlation separately from clinical CLISTER/DEFT2020 STS.

Gap 2: NER schemas conflate INN, brand name, and excipient under a single CHEM tag and do not annotate ATC codes or EMA administrative identifiers (QUAERO-D10, QUAERO-D12, MANTRAGSC-D14).

Recommendation: Define a regulatory NER schema with distinct INN, brand-name, excipient, ATC-code, EMA-identifier, and contraindication-qualifier classes. Re-annotate QUAERO EMEA and Mantra-GSC fr_emea (the only EMA-sourced subsets) using this schema, with regulatory affairs specialists as annotators.

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QUAERO-D10	Contient également: mannitol, hydroxyde de sodium. — Excipients tagged identically to INNs (CHEM), preventing entity sub-class distinction
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